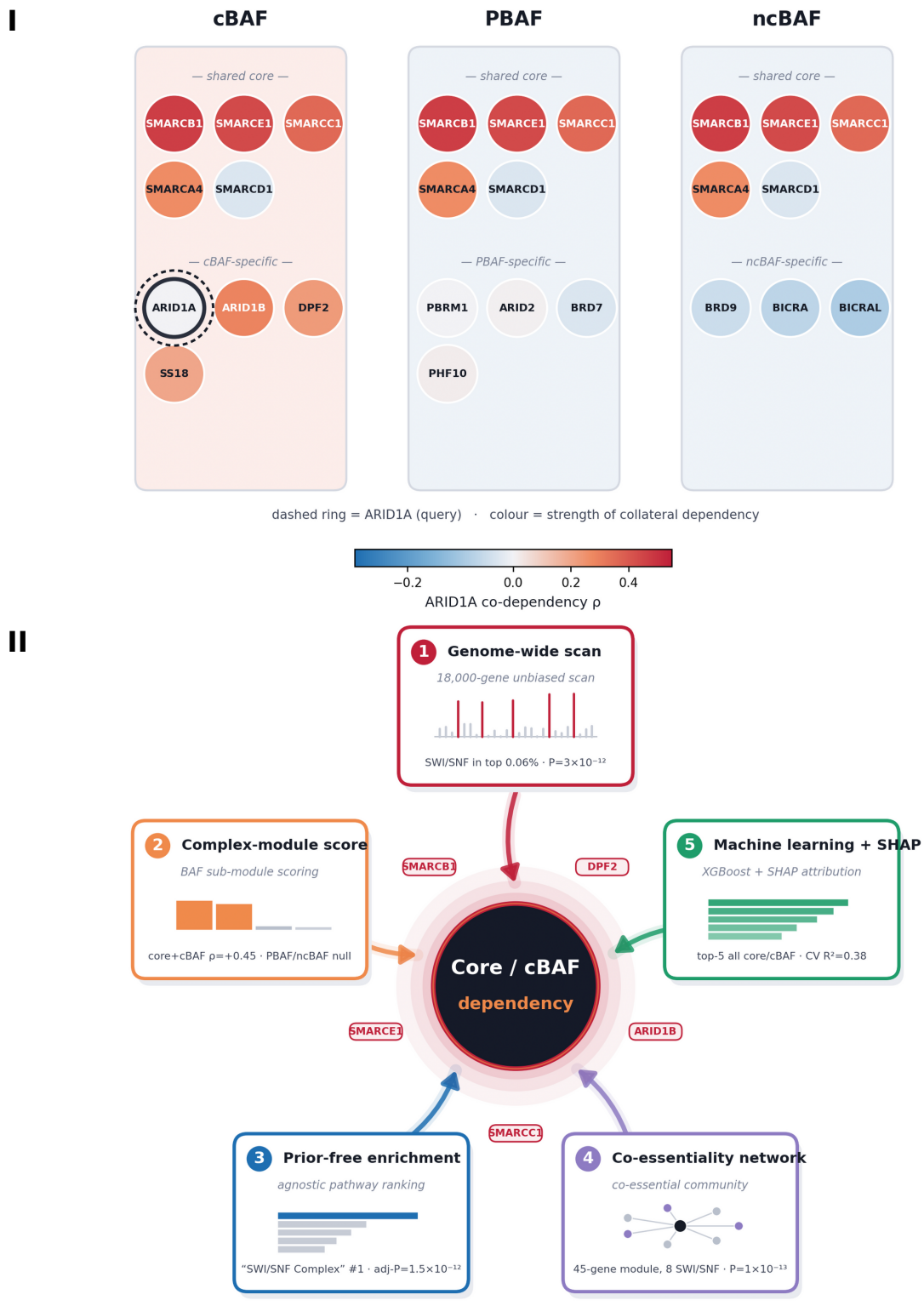
Supplementary material

Structure-informed CRISPR co-dependency maps the ARID1A-associated BAF module across cancer cell lines

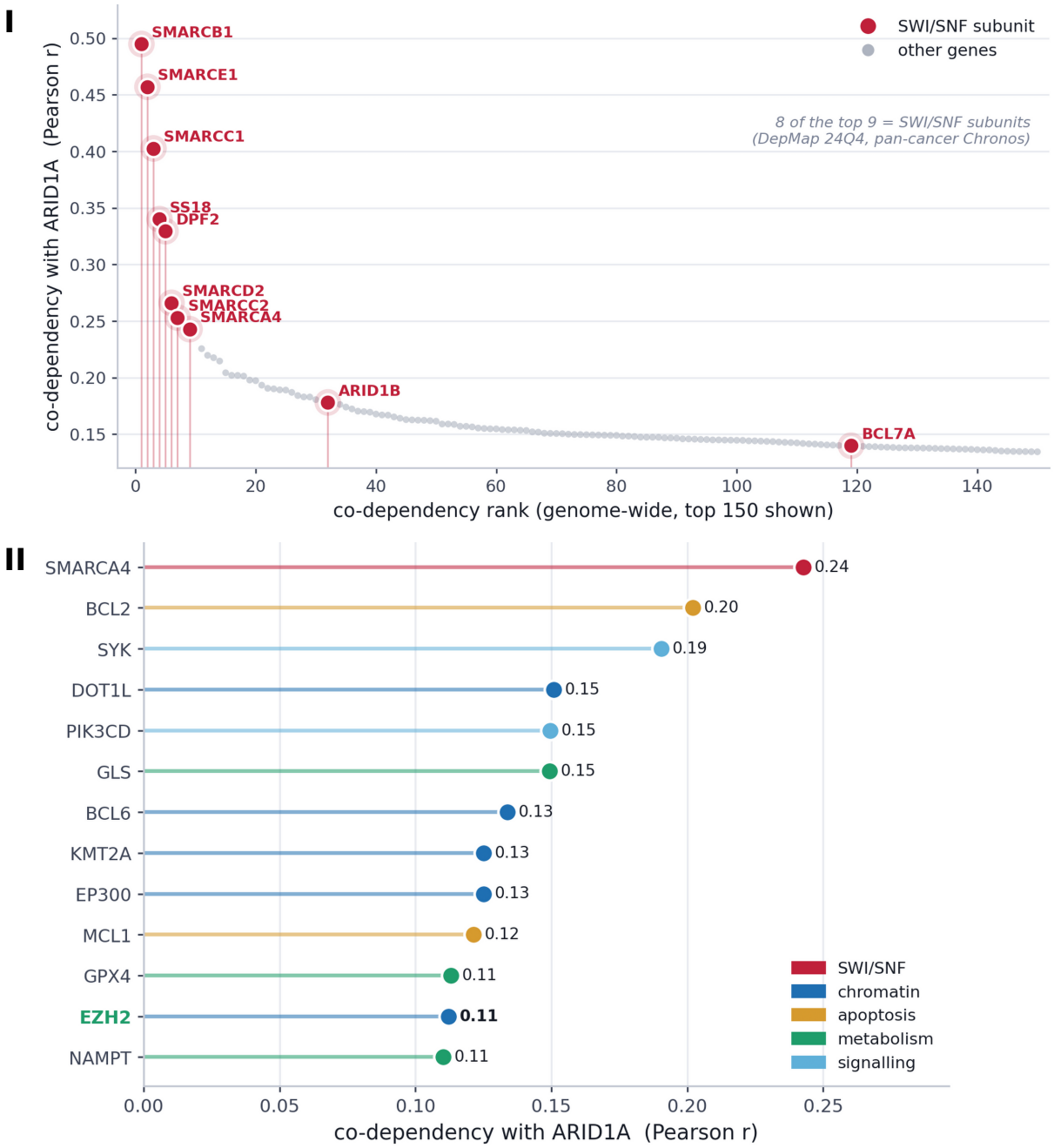
Zhiyong Tao, Juan Xiong, Yiwei Zhang, Chulin Zhong, Wenkai Gao, Guohua Fan

Supplementary Figure S1



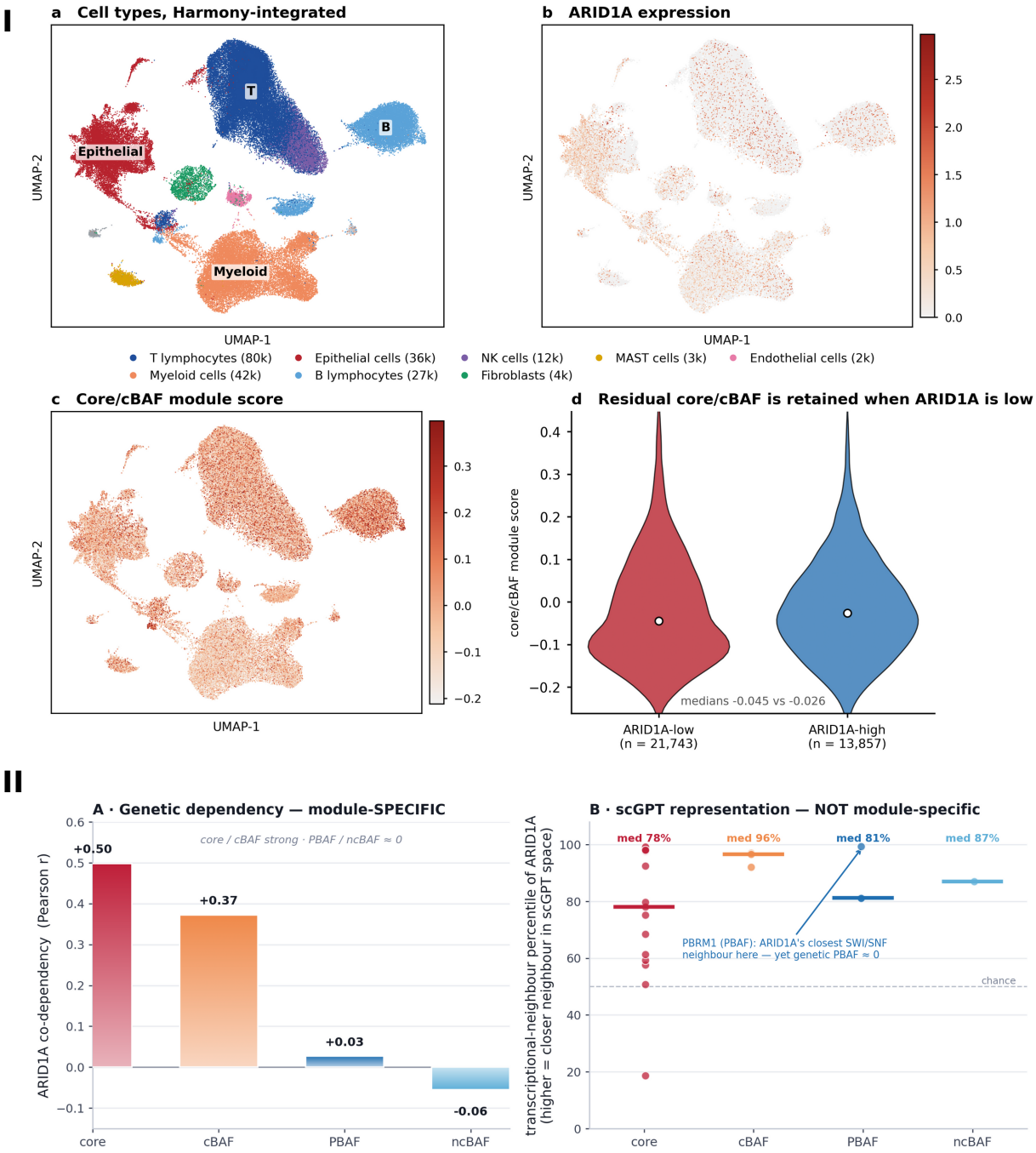
Sub-complex composition and the structure of the convergence argument. (I) The three SWI/SNF sub-complexes drawn by subunit membership. Each subunit is coloured by its pan-cancer co-dependency with ARID1A; ARID1A carries a dashed ring. (II) The five methodologically distinct analyses and the subunits each recovers. The analyses share one dependency matrix; four use no complex composition during discovery.

Supplementary Figure S2



Genome-wide ARID1A co-dependency landscape and the subset with existing chemical probes. (I) Top 150 genes ranked by pan-cancer co-dependency in DepMap 24Q4. (II) Co-dependent genes for which a chemical probe or clinical-stage inhibitor exists. These rankings describe co-dependency and do not establish genotype-selective efficacy.

Supplementary Figure S3



Single-cell expression context and foundation-model control. (I) GSE131907 cell types, ARID1A expression, core/cBAF module score and the descriptive cell-level split. Statistical inference in the main text uses 48 sample-level aggregates. (II) CRISPR co-dependency separates core/cBAF from PBAF/ncBAF, whereas the scGPT transcriptional representation places every module above chance and therefore does not resolve genetic specificity.